

Graphs - Set 4

Networks, flows, cuts

1. Carry out the complete critical path analysis for the project described below:

Activity	α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8	α_9	α_{10}	α_{11}
Time needed	6	2	10	1	4	2	7	7	9	2	4
Prerequisites	-	-	α_1	α_1	α_1	α_5	α_2	α_3	α_2	α_7	α_8
							α_4	α_6	α_4		α_{10}

2. Sketch the network whose vertices are s, a, b, c, d, t and whose arcs capacities are:

$(x, y) :$	(s, a)	(s, b)	(a, b)	(a, c)	(b, d)	(d, c)	(c, t)	(d, t)
$c(x, y) :$	5	3	3	3	5	2	6	2

Find a flow with value 7 and a cut with capacity 7. What is the value of a maximum flow, and why?

3. Let $D = (V, A)$ be a digraph, and suppose $\phi : A \rightarrow \mathbb{N}$ is a function, not necessarily a flow. Show that

$$\sum_{v \in V} \text{outflow}(v) = \sum_{v \in V} \text{inflow}(v).$$

Deduce that if s and t are the source and sink of a network, and ϕ is a flow, then

$$\text{outflow}(s) = \text{inflow}(t).$$